MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FIFTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG5DSESTA300– INTRODUCTION TO MULTIVARIATE ANALYSIS

Duration: 1Hour Maximum: 30 Marks

PRACTICAL EXAMINATION

Answer any five questions. Each question carries 6 marks.

Solve the questions using R in the computer lab. The R codes along with output must be written in the answer sheets.

1. An automotive researcher is evaluating whether vehicle efficiency and physical build characteristics differ systematically based on transmission type. Specifically, the researcher wants to test if the combined mean vector of Miles per Gallon (mpg) and Weight (wt) is significantly different between cars with automatic transmissions and those with manual transmissions. Using the built-in mtcars dataset in R, perform a Two-Sample Hotelling's T-squared Test to determine if there is a statistically significant multivariate difference between the two transmission types. [A] [1]

2. In the iris dataset, consider the two sepal measurements (Sepal.Length, Sepal.Width) as the first set of variables X, and the two petal measurements (Petal.Length, Petal.Width) as the second set of variables Y. Write the R code to compute the canonical correlations and extract the canonical coefficients for the both variable sets. [A] [1]

3. The built-in object eurodist contains road distances (km) between 21 European cities.
   (a) Obtain a two-dimensional configuration using classical MDS (cmdscale).
   (b) Plot the cities.
   (c) Compute the goodness-of-fit measures and the proportion of variation explained by each dimension. [A] [2]

4. Use the variables mpg, cyl, disp, hp, drat, wt and qsec of the mtcars data set.
   (a) Examine the correlation matrix and test its suitability for factor analysis.
   (b) Fit a two – factor maximum- likelihood model with varimax rotation. Report the loadings, uniqueness's and communalities.
   (c) Compute regression factor scores and plot the cars in the factor space. Which cars have the highest and lowest scores on Factor 1? [A] [3]

5. Two samples of sizes 70 and 100 are given from two normal populations $N_p(\mu_1,\Sigma)$ and $N_p(\mu_2,\Sigma)$.
   Let $\bar{X}^{(1)}=\begin{bmatrix}3846\\5040\\2940\end{bmatrix}$ $\bar{X}^{(2)}=\begin{bmatrix}6160\\8080\\5200\end{bmatrix}$ and S = $\begin{bmatrix}50910 & 8825 & 3925\\8825 & 11085 & 5070\\3925 & 5070 & 2710\end{bmatrix}$.
   Obtain the best discriminant function to discriminate between two populations and classify the observations X=$\begin{bmatrix}56\\78\\52\end{bmatrix}$. [A] [4]

6. For the distance matrix, apply single linkage and complete linkage methods and draw corresponding dendrograms. [A] [5]

$$\begin{bmatrix} 0 & 9 & 3 & 6 & 11 \\ 9 & 0 & 7 & 5 & 10 \\ 3 & 7 & 0 & 9 & 2 \\ 6 & 5 & 9 & 0 & 8 \\ 11 & 10 & 2 & 8 & 0 \end{bmatrix}$$

7. Use the knn () function of the package class (supplied with R) on the iris data.
   (a) Standardise the four measurements. Use 100 randomly selected flowers (set. seed (123)) as the training set and the remaining 50 as the test set. Classify the test flowers with k = 5 and give the confusion matrix and accuracy.
   (b) Choose k using leave-one-out cross-validation (knn.cv) for k = 1, …, 25 and plot the error against k. [A] [4]

8. Cross-classify the 32 cars of mtcars by number of cylinders (cyl) and number of forward gears (gear).
   (a) Construct the contingency table and test for association. Why does R give a warning, and how can it be avoided?
   (b) Perform correspondence analysis. How much of the inertia is explained by each dimension?
   (c) Draw and interpret the correspondence map. [A] [5]